KAVYA SUDHA

AWS Data Engineer

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PROFESSIONAL SUMMARY:

- 7+ years of practical experience as a Cloud Data Engineer, specializing in cloud data warehousing, big data processing, and ETL solutions for structured, semi-structured, and unstructured data.
- Extensive expertise in AWS services including RDS, IAM, EC2, Glue, Kinesis, Lambda, EMR, S3, SNS, SQS, Redshift, Athena, DynamoDB, Elastic Cache, Aurora, VPS, API Gateway, RDS, Step functions, CloudWatch, Cloud trial and so on.
- Experience in utilizing Azure platforms, employing Azure Data Factory, ADLS, Delta Lake, Synapse Analytics, and Databricks, and GCP BigQuery, Dataproc, Pub/Sub, and Dataflow to design scalable and efficient data solutions.
- Experience in building and architecting multiple data pipelines and end-to-end ETL and ELT processes for data ingestion and transformation, using PySpark for transforming large volumes of data efficiently.
- Experience in developing ETL applications on large volumes of data using different tools like MapReduce, Spark, and Scala.
- Working experience in data analysis techniques using Python libraries like NumPy, Pandas, SciPy, and visualization libraries of Python.
- Understanding of how to integrate data platforms utilizing Kafka producers and consumers, as well as how to perform pre-processing using storm topologies. Proficient in ingesting different types of data sources including RDBMS, files, streams, and APIs.
- Proven experience in designing and implementing scalable data warehousing systems with Snowflake and Teradata. Developed data models in BigQuery and GCS for optimized analytical querying.
- Worked with different Databases like Oracle, MySQL, SQL Server and was involved in schema design, writing complex SQL queries, and performance tuning for enhanced database efficiency.
- Hands-on experience with NoSQL databases, including Cassandra and MongoDB, for handling diverse business data requirements.
- Practical experience in coding MapReduce jobs on the Hadoop ecosystem, leveraging tools like Hive, Pig, HBase, Flume, Oozie, and Sqoop for advanced big data processing.
- In-depth knowledge of Hadoop architecture, including components such as HDFS, Job Tracker, Task Tracker, Name Node, Data Node, and the MapReduce programming paradigm.
- Competent in building RESTful APIs using Python for seamless data integration and real-time communication between enterprise systems.
- Adept at constructing interactive dashboards and visualizations using Tableau and Power BI, empowering stakeholders with actionable insights for informed decision-making.
- Expertise in implementing and automating CI/CD pipelines using Jenkins, Docker, and Kubernetes. Automated workflows with Python and other scripting languages to enhance deployment efficiency.
- Experienced in Agile methodologies, participating in sprints, daily stand-ups, and iterative development cycles, while also familiar with Waterfall approaches.

EDUCATION: Master's Degree in Information Systems and Technology from University of North Texas, Denton, Texas.

CERTIFICATIONS: AWS Certified Data Engineer - Associate

TECHNICAL SKILLS:

Cloud Technologies	AWS (EC2, ECS, ECR, S3, EMR, ELB, NLB, ALB, Redshift, Athena, Document DB, DynamoDB, Aurora, VPS, Glue, Kinesis, Lambda, SQS, SNS, API Gateway, RDS, Step functions, CloudWatch, Cloud trial, IAM) and Azure (Azure Data Factory, VMS, Cosmos

	DB, Functions, Blob storage, ADLS, Delta Lake, Synapse Analytics, and Databricks)and GCP (BigQuery, Dataflow, Dataproc, Pub/Sub, Composer, Cloud Storage).
Data Warehousing	BigQuery, Snowflake and Teradata.
Languages	Python, Scala, Spark
DevOps Tools	Azure DevOps, Docker, Terraform, Ansible, Jenkins
Data Visualization Tools	Power BI, Tableau, AWS Quick Sight.
Versioning Tools	GIT, GitHub, Bit-Bucket
Streaming Tools	AWS Kinesis, Apache Kafka, GCP Pub/Sub
SQL and NoSQL	MySQL, PostgreSQL, Oracle, SQL Server, MongoDB, Cassandra
File Types	Flat Files, Parquet, CSV, Avro, XML, JSON
Scripting	Shell, Python, PowerShell
Other Tools and Technologies	Informatica, JIRA, Splunk

WORK EXPERIENCE:
Agero Inc, Boston, MA
October 2023 to Till Now
Role: AWS Data Engineer
Responsibilities:

- Worked with AWS services such as EMR, EC2, S3, CloudFormation, and RedShift which provides fast and efficient processing of Big Data.
- Executed ETL workflows using AWS Glue to effectively extract, transform, and load data from various sources into Document DB, enhancing data integration procedures.
- Design and Develop ETL Processes in AWS Glue to migrate Campaign data from external sources like S3, ORC/Parquet/Text Files into AWS Redshift.
- Created external tables with partitions using AWS Athena and Redshift Used the AWS-CLI to suspend an AWS Lambda function processing an Amazon Kinesis stream, then resume it again
- Created and deployed data streaming solutions utilizing AWS Kinesis, allowing for the instantaneous processing and analysis of data.
- Migrated on-premises MySQL to AWS using Amazon RDS and DynamoDB Write scripts in Python to move data in JSON format from S3 buckets to Database tables.
- Implemented AWS EMR (Elastic MapReduce) to harness the capabilities of Hadoop and Spark, enabling streamlined processing and analysis of large-scale data.
- Implemented AWS SNS and SQS to incorporate real-time event processing and messaging functionalities.
- Processing data with Scala, spark, spark SQL and load in hive partition tables in parquet file format Develop spark job with partitioned RDD (like hash, range, custom) for faster processing.
- Implemented Spark Scripts using Scala, and Spark SQL to access hive tables into Spark for faster processing of data.
- Extensively worked on Spark using Scala on the cluster for computational (analytics), installed on top of Hadoop performed advanced analytical applications by making use of Spark.
- Create and Maintain Tables and views in Snowflake Measured Efficiency of Hadoop/Hive environment ensuring SLA is met Optimized the TensorFlow Model for efficiency.
- Developed custom scripts and tools employing Oracle's PL/SQL language to automate data validation, cleansing, and transformation processes.
- Deploy EC2 instances for the Oracle database
- Developed and executed a migration strategy to move Data Warehouse from an Oracle platform to AWS Redshift.
- Developed Automation Regressing Scripts for validation of ETL process between multiple databases like AWS Redshift, Oracle, MongoDB, T-SQL, and SQL Server using Python
- Collect the data using Spark streaming and dump it into HBase and Cassandra.

- Used the Spark-Cassandra Connector to load data to and from Cassandra.
- Implemented test scripts to support test-driven development and continuous integration.
- Developed Data Cleansing and Transformational workflows from multiple data sources.

Molina Healthcare, Long Beach, CA
December 2021 to September 2023
Role: Azure Data Engineer
Responsibilities:

- Implementing the various Azure Services such as securing storage in Azure: Blob, Databricks SQL configuration, configuring Azure AD, Configuring AD Connect, ADFS, and implementing Security in Azure: Key Vault, Azure Active Directory & RBAC.
- Worked on the Azure Databricks implementation with PySpark with Databricks SQL followed with Medallion architecture & storing the data in the Azure Blob storage in 3 layers Bronze, Silver, and Gold Layers.
- Worked on creating the Azure ADF Pipelines and Azure SQL Server DB.
- Worked on Data migration from DataStage into the ADF with the Azure Data Factory Pipeline creation.
- Integrated Azure DevOps tool with Planning & CI/CD Processes.
- Worked on the Databricks services, ADF, ADLS, AAD, Pipelines, Azure SQL Server DB, and Azure functions.
- Worked on the Cloud Native tools, Azure DevOps Automation process with Azure Pipelines and GitHub.
- Developed both normal and materialized views for reporting purposes, tailored to specific business scenarios.
- Developed data pipelines to source data from PostgreSQL databases and REST APIs, ensuring efficient data ingestion.
- Developed Azure Data Factory pipelines to orchestrate multiple Hive, Pig, Azure Data Lake, Cosmos DB, Git, Sqoop, and Spark jobs.
- Integrated data from MongoDB, MS SQL, and cloud sources using Azure Data Factory, SQL API, and MongoDB API.
- Built PySpark code to cleanse and transform data for downstream data science applications.
- Created a framework in PySpark to dynamically parse JSON files and insert the processed data into specific database tables.
- Implemented generic code to dynamically handle nested JSON data parsing.

BBVA, Birmingham, AL
June 2019 to November 2021
Role: Data Engineer
Responsibilities:

- Developed and deployed Spark application using Pyspark to compute popularity score for all the contents using an algorithm and load the data into Elastic Search for the App content management team to consume.
- Converted the data science model developed by data scientists from Python to Pyspark and operationalized the model using Python and shell scripts to automate the process of running the model on new data as required and save the results to final Phoenix tables
- Migrate an in-house database to AWS Cloud and designed, built, and deployed a multitude of applications utilizing the AWS stack (Including S3, EC2, RDS, Redshift, and Athena) by focusing on high-availability and auto-scaling.
- Used Apache Kafka to aggregate web log data from multiple servers and make them available in Downstream systems for analysis and used Kafka Streams to Configure Spark streaming to get information and then store it in HDFS.
- Configured Spark streaming to receive Kafka input streams from the Kafka and Specified exact block interval for data Processing into HDFS using Scala.
- Created monitors, alarms, notifications, and logs for Lambda functions, Glue Jobs, and EC2 hosts using CloudWatch and used AWS Glue for the data transformation, validation, and data cleansing.
- Worked with cloud-based technology like Redshift, S3, AWS, EC2 Machine, etc. Extract the data from the Oracle Financials and the Redshift database. Create Glue jobs in AWS and load incremental data to the S3 staging area and persistence area.
- Handled importing data from various data sources, performed transformations using Hive, MapReduce, and loaded data into HDFS.
- Deployed applications using Jenkins's framework integrating Git- version control with it.
- Improving the performance and optimization of the existing algorithms in Hadoop using Spark Context, Spark-SQL, Data Frame, Pair RDDs, and YARN.

- Created Data pipelines for Kafka cluster and processed the data by using spark streaming and worked on streaming data to consume data from Kafka topics and load the data to the landing area for reporting in near real-time.
- Scheduled Airflow DAGs to run multiple Hive and Pig jobs, which independently run with time and data availability, and Performed Exploratory Data Analysis and Data visualization using Python, and Tableau.

Homesite Insurance, Boston, MA
June 2017 to May 2019
Role: Big Data Developer
Responsibilities:

- Built scalable and secure data pipelines for large datasets.
- Gathered requirements for new data source ingestion, including lifecycle, data quality checks, transformations, and metadata enrichment.
- Created data pipelines utilizing Flume and Sqoop to ingest customer behavioral data into HDFS for comprehensive analysis.
- Utilized prominent big data analytics tools, such as Hive and MapReduce, to perform insightful analysis on Hadoop clusters.
- Supported data quality management by implementing proper checks in data pipelines.
- Improved Data Ingestion Framework by enhancing data pipelines' robustness and security.
- Implemented data streaming using Kafka and Talend for multiple data sources.
- Supported Map Reduce Programs that are running on the cluster.
- Monitoring the jobs to analyze performance statistics.
- Worked hands-on with No-SQL databases like Cassandra and MongoDB for POC purposes in storing images and URIs.
- Managing and scheduling batch Jobs on a Hadoop Cluster using Oozie.
- Written Hive queries to process the data for visualization.
- Supported in setting up the QA environment and updating configurations for implementing scripts with Pig and Sqoop.